

Cesar Peralta

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Education

McGill University – Montréal

GPA: 3.72

B.Sc., Major in Computer Science (Artificial Intelligence), Minor: Entrepreneurship | Expected May 2028

- Relevant Courses: Algorithms and Data Structures, Applied Machine Learning

Technical Skills

- **Languages:** Python, Java, C, SQL
- **Frameworks & Tools:** Git/GitHub, Linux, Docker, pandas, LangGraph, FastMCP, APIs, ChromaDB, PostgreSQL, Ollama, Agent Skills, RAG, NLP, CSV, JSON, ETL
- **Spoken languages:** English and Spanish (native/bilingual), French (full professional)

Experience

Clinical Data Engineer / Clinical Research Coordinator

Lady Davis Institute for Medical Research | March 2024 – Present

- Automated approximately 90% of an image processing workflow (previously required about 7 hours/week), using **Python** to control a desktop application, validate conditions, export data, and generate PDF reports.
- Achieved a 98% reduction in preparation time in patient email delivery through a **Python** rule-based patient report automation that selects context-specific templates while eliminating manual errors.
- Built **Python** ETL data pipelines to batch-upload over 6,000 patient records to the clinical database.

Undergraduate Researcher

McGill School of Computer Science | May 2026 – July 2026

- Reproduced a published machine learning benchmark for protein-interaction prediction to establish a 64.9% baseline before data-quality experiments.
- Ran 13 dataset variants and 90 repeated experiments on a compute cluster; filtering low-confidence training examples raised accuracy from 63.6% to 75.4%.
- Built scalable preprocessing and data leakage checks across two protein-interaction datasets, coordinating parallel experiments with **Git** branches and worktrees.

Selected Projects

MCP Clinical Agent

- Built a local AI agent in **Python** using **LangGraph** and **FastMCP** that converts natural language clinical questions into **PostgreSQL** queries, running in **Docker** with Ollama local inference, automatic **SQL** error recovery, conversational memory, and CSV exports (achieved 20/20 on benchmark runs).

RAG Loremaster

- Built a retrieval-augmented generation pipeline in **Python** using **ChromaDB** and **Docker** to crawl web wikis and synthesize cited answers via Ollama and Gemini APIs, scoring 81% accuracy on evaluation.

Clinical Medication Parser

- Built a two-stage local LLM pipeline using **Python**, **Ollama**, **Pydantic**, and **pandas** that extracts medications from unstructured clinical notes into validated structured data with source text evidence. Achieved approximately 87% case-level quality across 30 cases.